SUPPORTING ADULT GUIDANCE

Which is larger?

Place Value · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Column Race, two numbers sit in a place-value chart lined up on the decimal point. Learners predict which is larger, then press Compare next column; columns light up from the left, green when the digits match and gold at the first column that differs, and empty columns show a grey 0. Eight pairs include 0.5 and 0.45, 2,099 and 2,109, and 0.3 and 0.300. Six questions follow, and a stretch explains why the first different column decides. Learners then compare pairs with >, = and <, order sets of four using a chart, and make numbers that fit between two others.

Driving Question: *Is a longer decimal always larger?*

The mathematics, in plain terms

Comparing by place value works because each column is worth more than all the columns to its right combined (for example 0.09 + 0.009 + … < 0.1). The common errors are treating the digits after the point as a whole number ("45 is more than 5, so 0.45 > 0.5") or assuming more digits means larger. Lining the numbers up on the point, and padding with zeros if it helps, removes both.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — 0.5 or 0.45?

Which is larger, 0.5 or 0.45?

0.5

Ones		Tenths	Hundredths
0	•	5	

0.45

Ones		Tenths	Hundredths
0	•	4	5

Lined up on the point. The ones match (0 and 0); the tenths differ: 5 against 4.

1. Ones: 0 and 0.	the same
2. Tenths: 5 and 4.	different: 5 tenths is more
3. The 5 hundredths in 0.45 cannot make up a whole tenth.	0.5 is larger

0.5 > 0.45. The longer decimal is the smaller one here.

Example 2 — 2,099 or 2,109?

Which is larger, 2,099 or 2,109?

2,099

Thousands	Hundreds	Tens	Ones
2	0	9	9

2,109

Thousands	Hundreds	Tens	Ones
2	1	0	9

The thousands match; the hundreds decide: 1 hundred beats 0 hundreds.

1. Thousands: 2 and 2.	the same
2. Hundreds: 0 and 1.	different: 1 hundred is more
3. The 99 in 2,099 is less than one hundred.	2,109 is larger

2,109 > 2,099, even though 2,099 has bigger digits at the end.

Predict, compare column by column, then explain

Predicting first exposes the "longer is larger" belief; the column-by-column reveal then tests it. The stretch gives the reason. Quick comparisons, ordering sets and fitting numbers between others then use the method in increasingly open ways.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Column Race and six questions. Stretch: why does the first different column decide?	~16 min
Movement	Quick compare: >, = or <.	~7 min
Reflection	The sorter: order four decimals, with the chart beside them.	~8 min
Creativity	Fit it between: make numbers between two others.	~10 min
Rest	The first difference is the one that counts.	~2 min
Nutrition	Intellectual: judgement.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a longer decimal is larger (0.45 > 0.5).
Ask: “How many tenths does each have?”

Reading the digits after the point as a whole number (45 is more than 5).
Ask: “Is the 4 in 0.45 worth 4, or 4 tenths?”

Thinking a shorter decimal is always smaller, after correcting the first error.
Ask: “Compare 0.5 and 0.55. What happens now?”

Lining numbers up on the right-hand edge.
Ask: “Where is the point in each number? Are the ones in the same column?”

Thinking 0.3 and 0.300 are different.
Ask: “What is in the hundredths and thousandths columns of 0.3?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: tenths, hundredths and thousandths (Lesson 4). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner orders 3.7, 3.07, 3.17 and 3.71 from smallest as 3.7, 3.07, 3.17, 3.71, reading the digits after the point as whole numbers (7, 7, 17, 71). The aim is a question that brings the columns back.

- “Can you write them one above the other with the points lined up?”

- “Which column do you compare after the ones?”
- “Which have 7 tenths? Which have 0 or 1?”
- “Now, of the two with 7 tenths, which is larger?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners add zeros to make the decimals the same length?
It can help, as long as they understand why the zeros change nothing (Lesson 3). The aim is to compare columns, with or without them.

Why include whole numbers?
The method is the same, and the pair 2,099 and 2,109 shows that the first different column decides, not the last digits.

Is the stretch required?
No. The secure core is comparing and ordering decimals reliably and explaining why length does not decide.

What comes next?
Lesson 7: rounding, which is nearest?

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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